

Zhihao Zhu

Postdoctoral Fellow, Hong Kong University of Science and Technology (HKUST)

Email: zhihaozhu@ust.hk | Website: <https://zzh9568.github.io> | GitHub: zzh9568

Google Scholar: <https://scholar.google.com/citations?user=745H-yoAAAAJ> | ORCID: 0000-0001-5814-1939

RESEARCH INTERESTS

Trustworthy AI; large language models; LLM privacy; training data detection; data revocation; membership inference; recommender systems; model stealing.

EDUCATION

Ph.D. in Computer Science, University of Science and Technology of China 2025

Advisor: Prof. Defu Lian

Bachelor's Degree, Fuzhou University 2020

ACADEMIC APPOINTMENT

Postdoctoral Fellow, Hong Kong University of Science and Technology 2025-Present

Advisor: Prof. Yi Yang

SELECTED PUBLICATIONS

- **Zhihao Zhu**, Yi Yang, Yangyang Fan, Defu Lian. *Forget Me If You Can: Auditing User Data Revocation in Recommendation Systems*. Information Systems Research, 2026.
- **Zhihao Zhu**, Yi Yang, Defu Lian. *TDDBench: A Benchmark for Training Data Detection*. International Conference on Learning Representations (ICLR), 2025.
- Hongyi Tang*, **Zhihao Zhu***, Yi Yang. *Identifying Pre-training Data in LLMs: A Neuron Activation-Based Detection Framework*. Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025. *Equal contribution.
- **Zhihao Zhu**, Chenwang Wu, Rui Fan, Defu Lian, Enhong Chen. *Membership Inference Attacks against Sequential Recommender Systems*. ACM Web Conference (WWW), 2023.
- **Zhihao Zhu**, Chenwang Wu, Min Zhou, Hao Liao, Defu Lian, Enhong Chen. *Resisting Graph Adversarial Attack via Cooperative Homophilous Augmentation*. ECML PKDD, 2022.
- Defu Lian, **Zhihao Zhu**, Kai Zheng, Yong Ge, Xing Xie, Enhong Chen. *Network Representation Lightening from Hashing to Quantization*. IEEE Transactions on Knowledge and Data Engineering, 2022.

WORKING PAPERS

- **Zhihao Zhu**, Jiale Han, Yi Yang. *HoneyImage: Verifiable, Harmless, and Stealthy Dataset Ownership Verification for Image Models*. Under review at MISQ.
- **Zhihao Zhu**, Yi Yang, Chenwang Wu, Defu Lian. *GraphMSA: Stress Testing Graph Classification Services Against Model Stealing Attacks*. Under review at IJOC.
- **Zhihao Zhu**, Hongyi Tang, Yi Yang, Ahmed Abbasi. *Revealing Training Data Exposure in Vision-Language Large Models via Parameter Gradients*. Under review at Nature Communications.

- **Zhihao Zhu**, Rui Fan, Chenwang Wu, Yi Yang, Defu Lian, Enhong Chen. *Model Stealing Attacks against Recommender Systems*. Under review at IEEE Transactions on Dependable and Secure Computing.
- **Zhihao Zhu**, Yi Yang. *RecShield: Output-Level Attribute Unlearning in Recommender Systems*. Under review at Information Systems Research.
- **Zhihao Zhu**, Ninglu Shao, Defu Lian, Chenwang Wu, Zheng Liu, Yi Yang, Enhong Chen. *Understanding Privacy Risks of Embeddings Induced by Large Language Models*.

AWARDS

- **National Graduate Scholarship (China)**, 2022.
- **KDD Cup 2023 "Next Product Generation" Challenge**, 2nd Place.